

Aditya Sugandhi

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SUMMARY

Full-Stack Software Engineer with 5+ years building, shipping, and maintaining production full-stack applications across healthcare, finance, and research domains. Expert in Python (FastAPI, Flask), TypeScript/JavaScript (Node.js, Next.js), and Java (Spring Boot) for backend development, with React, Angular, and Vue.js for frontend applications. Strong command of SQL and relational database design—architected PostgreSQL and MySQL schemas with normalization, foreign key relationships, and horizontal sharding for 500M+ record databases; wrote complex SQL queries with multi-table joins, CTEs, window functions, and aggregations improving performance by 2-3x. Holds M.S. in Computer Science from Florida State University. Proven ability to articulate complex technical concepts clearly to both technical and non-technical stakeholders, translating database architecture, API design, and distributed systems into actionable business insights for cross-functional teams.

EDUCATION

Florida State University

Master of Science in Computer Science (Advanced Degree)

Aug 2022 – Dec 2023
Tallahassee, FL

- Advanced degree in Computer Science with relevant technical focus in Distributed Systems, Advanced Database Systems, Machine Learning, and Data Communications

SRM Institute of Science & Technology

Bachelor of Technology in Computer Science

Jun 2015 – May 2019
Chennai, India

WORK EXPERIENCE

Software Engineer II

Florida State University — Finance Technology Division

Nov 2024 – Present
Tallahassee, FL

- Built, shipped, and maintained production full-stack applications serving 10K+ users with Python (FastAPI) and Java (Spring Boot) backend microservices, React and Angular frontend, and PostgreSQL relational database, processing 5M+ daily requests with 99.99% uptime through distributed architecture.
- Designed and queried PostgreSQL relational database schemas with normalization, foreign key relationships, and horizontal sharding for 500M+ records; wrote complex SQL queries with multi-table joins, CTEs, window functions, and aggregations for time-series analytics and compliance reporting, improving query performance by 2x through strategic indexing and query optimization.
- Developed 37 RESTful API endpoints using Python (FastAPI) and Java (Spring Boot) with real-time WebSocket streaming; implemented distributed caching with Redis reducing API latency by 45% (850ms to 470ms) and enabling sub-second response times for real-time dashboards serving concurrent users.
- Articulated complex technical concepts clearly to both technical and non-technical stakeholders, translating database schema design, SQL query optimization strategies, API performance metrics, microservices architecture, and infrastructure costs into business impact through presentations and documentation; enabled data-driven decision making for finance and compliance teams.
- Maintained production systems end-to-end including monitoring, reliability, and incident response; established observability infrastructure with Prometheus/Grafana, implemented automated alerting, and reduced MTTR by 60% through systematic troubleshooting and root cause analysis.
- Led technical initiatives mentoring 6 engineers on full-stack development patterns, SQL optimization, and API design best practices; conducted code reviews and set team standards improving development velocity by 25%.

Machine Learning Engineer

Prof. Olmo Zavala Romero — FSU

Jan 2023 – Aug 2024
Tallahassee, FL

- Built and shipped full-stack data applications processing 1TB+ datasets with Python (FastAPI) backend and React frontend; designed PostgreSQL relational database schemas with normalized tables and wrote advanced SQL queries with CTEs, subqueries, and complex joins for analytics achieving 99% job success rate.
- Shipped production ML infrastructure on Kubernetes with RESTful APIs built in Python and real-time monitoring dashboards; maintained system reliability through automated testing, performance tuning, and cost optimization reducing cloud costs by 40%.
- Communicated technical architecture decisions clearly to non-technical research stakeholders, explaining database design trade-offs, SQL query performance, API architecture, distributed systems design, and infrastructure costs through technical presentations; delivered data-driven insights supporting strategic research decisions.

Software Engineer I

Aspire Systems

Oct 2020 – Jul 2022
Chennai, India

- Developed, shipped, and maintained production full-stack applications using Java (Spring Boot) backend, React and Angular frontend, and PostgreSQL relational database processing 500K+ events/day; shipped features for financial reporting platform serving enterprise clients with 99.5% availability.
- Designed and queried PostgreSQL relational database schemas with normalization and foreign key relationships; wrote complex SQL queries with multi-table joins, aggregations, and subqueries for general ledger integration and P&L calculations; implemented database indexing strategies and query optimization improving report generation speed by 3x.
- Built RESTful APIs and microservices with Java (Spring Boot) and Kafka event streaming; implemented distributed tracing and monitoring with Prometheus/Grafana ensuring production system reliability and performance under high load.
- Communicated technical concepts clearly to product managers and business stakeholders, explaining API design decisions, database performance metrics, SQL optimization strategies, and system architecture trade-offs through technical presentations and documentation; influenced product roadmap through clear articulation of technical constraints and opportunities.

- Mentored **5 junior engineers** on full-stack development, SQL best practices, and API design patterns; led code reviews and established team standards improving code quality and development velocity.

Associate Software Engineer
IMPETUS Technologies

Jun 2019 – Sep 2020
Noida, India

- **Developed and maintained full-stack applications** with **Java backend** and **JavaScript frontend**; built **RESTful APIs** handling **100K+ daily requests** with **MySQL relational database integration** and Redis caching for low-latency data access.
- Maintained production systems including **SQL query optimization**, API performance tuning, and bug fixes; implemented workflow automation improving engineering team productivity and system reliability.

PROJECTS

- **BookedFlow Slack Bot (Full-Stack Integration)** — Deployed and maintained production Slack bot with **TypeScript/Node.js backend** and Socket Mode API achieving **24/7 availability**. Built **WebSocket connections**, OAuth 2.0 authentication, and **RESTful APIs** on AWS EC2. Implemented real-time messaging with **sub-40-second response times**. Tech: **TypeScript, Node.js, Next.js, WebSocket, REST APIs, AWS**.
- **Building Information Portal (Full-Stack)** — Built and shipped production web application managing **14.6M+ sq. ft.** of university infrastructure with **Angular frontend** and **Java (Spring Boot) backend**. Designed **PostgreSQL relational database schema** with normalized tables and **role-based access control**. Built **multi-criteria search with complex SQL queries** (joins, aggregations) serving **50K+ users**. Tech: **Java, Spring Boot, Angular, PostgreSQL, SQL, REST APIs**.
- **SkillSync MCP (Full-Stack Security Tool)** — Built, shipped, and published **npm package** with **TypeScript backend (3,085 LOC)** providing security-gated skill management for **8 AI IDE platforms**. Developed automated security scanning detecting **60+ threat patterns**. Integrated with GitHub API and **SQLite relational database** for skill metadata storage with SQL queries. Tech: **TypeScript, Node.js, SQLite, SQL, REST APIs, npm**.

TECHNICAL SKILLS

- **Full-Stack Development:** 5+ Years Building, Shipping, and Maintaining Production Full-Stack Applications; End-to-End Application Development; API Design; Database Architecture; Authentication/Authorization; Production Deployment; System Maintenance
- **Backend Technologies (Proficient):** Python (FastAPI, Flask), TypeScript/JavaScript (Node.js, Next.js), Java (Spring Boot), RESTful APIs, WebSocket, Microservices Architecture, Event-Driven Architecture
- **Frontend Technologies (Proficient):** React, Vue.js, Angular, TypeScript, JavaScript, HTML/CSS, Responsive Design, Real-Time Dashboards
- **SQL & Relational Databases (Strong Command):** PostgreSQL (Schema Design, Normalization, Query Optimization, Indexing, Sharding, Complex Joins, CTEs, Window Functions, Aggregations), MySQL (Schema Design, Query Optimization), SQL (Multi-Table Joins, Subqueries, Performance Tuning), Database Modeling, Foreign Keys, Transaction Management
- **Cloud & Infrastructure:** AWS (EKS, Lambda, S3, EC2), Kubernetes, Docker, Distributed Systems, CI/CD Pipelines
- **Data Processing:** Apache Kafka, Apache Spark, Apache Airflow, ETL/ELT Pipelines, Batch & Streaming Processing, Redis Caching
- **Monitoring & Reliability:** Prometheus, Grafana, Distributed Tracing, Observability, Production Support, Incident Response, Performance Tuning
- **Communication & Leadership:** Articulating Complex Technical Concepts Clearly to Both Technical and Non-Technical Audiences, Technical Presentations, Stakeholder Management, Cross-Functional Collaboration, Mentoring, Agile/Scrum
- **Additional:** MongoDB, Time-Series Databases (InfluxDB), gRPC, Code Reviews, Testing (Unit, Integration, E2E)

PUBLICATIONS & CERTIFICATIONS

- Review Classification & False Feedback Detection (IJAST) | AWS Solutions Architect (Udemy, 2024)